COURSE 1 · MATHEMATICS FOUNDATIONS

Mathematics foundations

Visual companion and Symbols Sheets

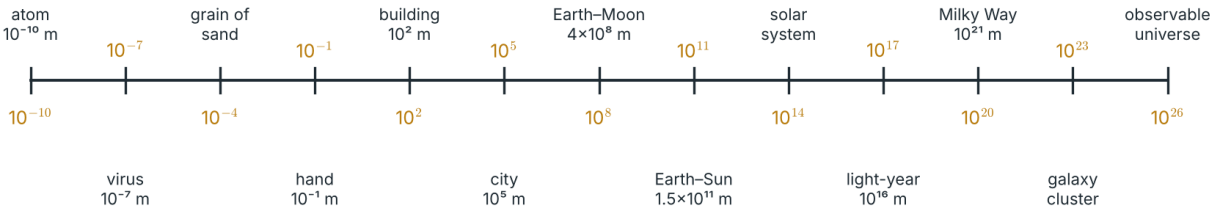
One section per module: the Symbols Sheet, with every numbered equation from the audio reader in ordinary notation, then the module's key diagram.

Keep it open while you listen, and beside you for the workbook.

Module 1.1 · Symbols and diagrams

Numbers and arithmetic fluency

1.1-1	Percentage change	$\%\ \text{change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100$
1.1-2	Multiplying powers	$(a \times 10^m)(b \times 10^n) = ab \times 10^{m+n}$
1.1-3	Dividing powers	$\frac{a \times 10^m}{b \times 10^n} = \frac{a}{b} \times 10^{m-n}$
1.1-4	Negative and zero powers	$10^{-n} = \frac{1}{10^n}, \qquad 10^0 = 1$



Each tick is a factor of 1000. Scientific notation lets you move along this line by adding and subtracting exponents.

Powers of ten from atoms to the observable universe. Scientific notation turns multiplication of huge or tiny numbers into addition of exponents.

Prefix	Symbol	Power of ten	Example
giga	G	10^9	gigawatt power station
mega	M	10^6	megajoule
kilo	k	10^3	kilogram
milli	m	10^{-3}	millimetre
micro	μ	10^{-6}	micrometre (cell size)
nano	n	10^{-9}	nanometre (molecules)

Module 1.2 · Symbols and diagrams

Units, measurement and dimensional analysis

1.2-1	The newton in base units	$1\ \mathrm{N} = 1\ \mathrm{kg\,m\,s^{-2}}$
1.2-2	Area and volume conversions	$1\ \mathrm{m^2} = 10^4\ \mathrm{cm^2}, \qquad 1\ \mathrm{m^3} = 10^6\ \mathrm{cm^3} = 1000\ \mathrm{L}$
1.2-3	Dimensional argument for a pendulum	$[T] = \sqrt{\frac{[L]}{[g]}} = \sqrt{\frac{\mathrm{m}}{\mathrm{m\,s^{-2}}}} = \mathrm{s}$
1.2-4	Percentage uncertainty	$\%\ \text{uncertainty} = \frac{\Delta x}{x} \times 100$

72 km/h × 1000 m/km × 1 h/3600 s = 20 m/s

km cancels h cancels

A conversion factor is a fraction equal to 1. Multiply by as many as you need, arranged so unwanted units cancel.

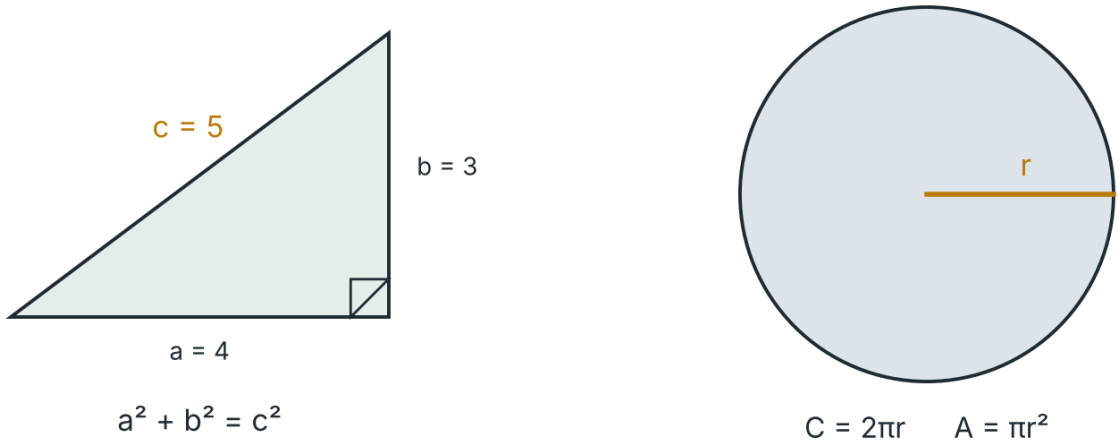
Chaining conversion factors. Each factor equals one, so the quantity is unchanged while the units are swapped.

Quantity	SI unit	Symbol	In base units
force	newton	N	$\mathrm{kg\,m\,s^{-2}}$
energy	joule	J	$\mathrm{kg\,m^2\,s^{-2}}$
power	watt	W	$\mathrm{J\,s^{-1}}$
pressure	pascal	Pa	$\mathrm{N\,m^{-2}}$
charge	coulomb	C	A s
frequency	hertz	Hz	$\mathrm{s^{-1}}$

Module 1.3 · Symbols and diagrams

Geometry

1.3-1	Pythagoras' theorem	$a^2 + b^2 = c^2$
1.3-2	Diagonal of a box	$d = \sqrt{a^2 + b^2 + c^2}$
1.3-3	Scaling of length, area and volume	$L \to kL, \quad A \to k^2 A, \quad V \to k^3 V$
1.3-4	Circle	$C = 2\pi r, \qquad A = \pi r^2$
1.3-5	Sphere	$A = 4\pi r^2, \qquad V = \frac{4}{3}\pi r^3$

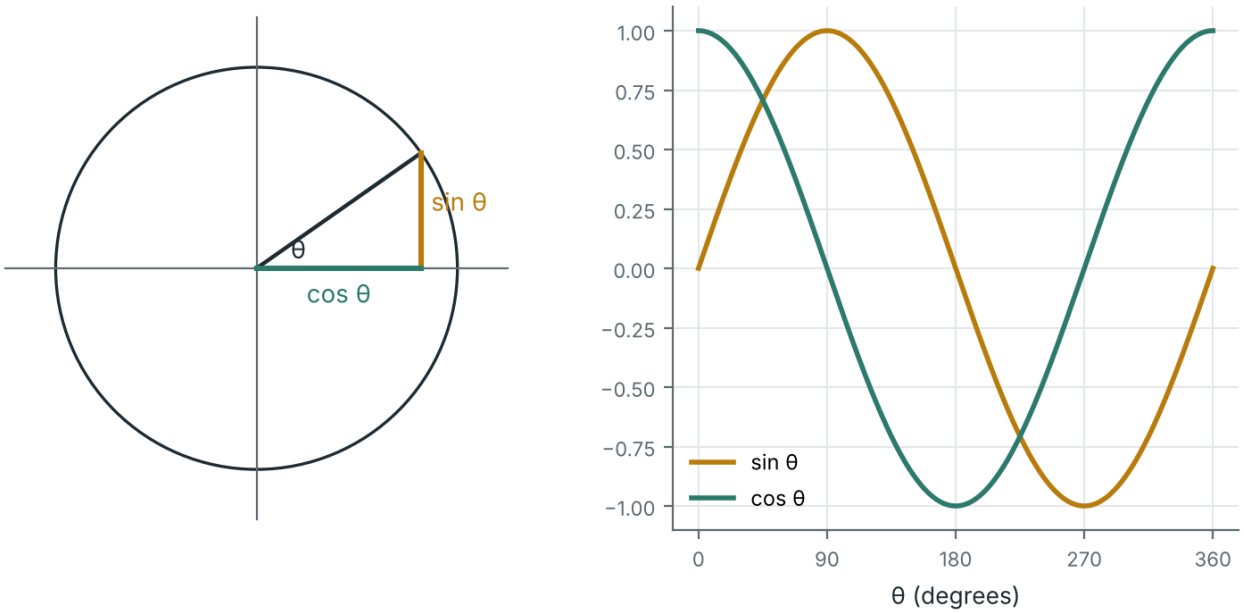


Pythagoras in a 3-4-5 triangle, and the circle formulas.

Module 1.4 · Symbols and diagrams

Trigonometry

1.4-1	Trigonometric ratios	$\sin\theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos\theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan\theta = \frac{\text{opp}}{\text{adj}}$
1.4-2	Resolving a vector of size F at angle θ	$F_x = F\cos\theta, \qquad F_y = F\sin\theta$
1.4-3	Degrees and radians	$\pi \text{ rad} = 180^\circ$
1.4-4	Arc length	$s = r\theta$
1.4-5	Pythagorean identity	$\sin^2\theta + \cos^2\theta = 1$
1.4-6	General sinusoid	$y = A\sin(\omega t + \phi), \qquad T = \frac{2\pi}{\omega}$



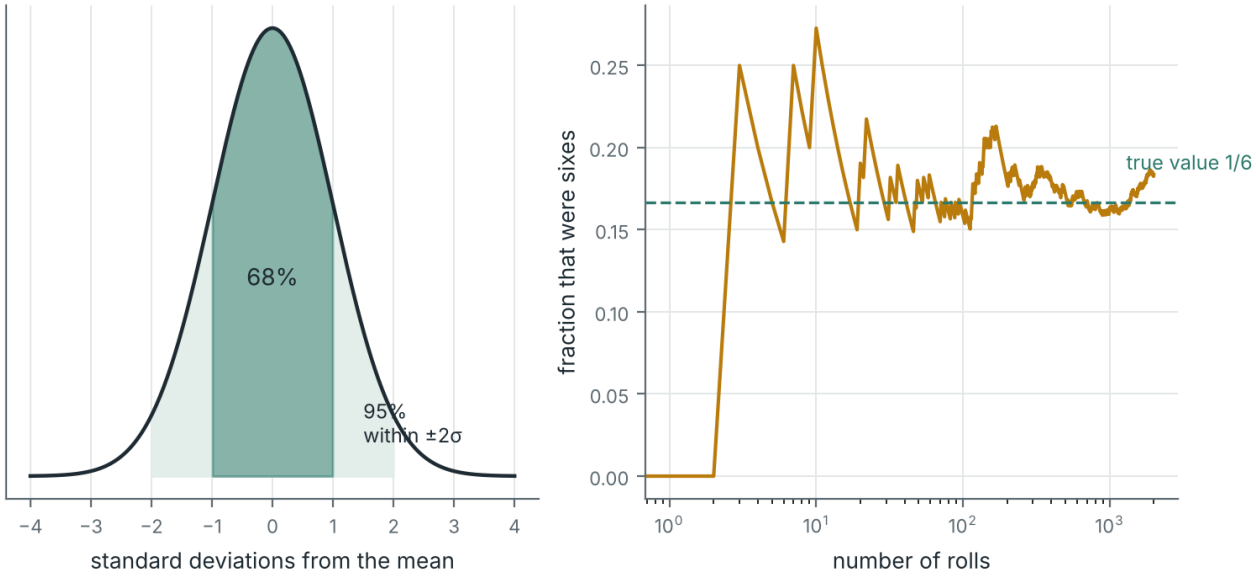
Left: on the unit circle, cos θ is the horizontal coordinate and sin θ the vertical. Right: sweeping θ round the circle traces the sine and cosine waves.

Angle	Radians	sin	cos	tan
0°	0	0	1	0
30°	π/6	1/2	√3/2	1/√3
45°	π/4	1/√2	1/√2	1
60°	π/3	√3/2	1/2	√3
90°	π/2	1	0	undefined

Module 1.5 · Symbols and diagrams

Probability and statistics

1.5-1	Addition rule (mutually exclusive)	$P(A \text{ or } B) = P(A) + P(B)$
1.5-2	Multiplication rule (independent)	$P(A \text{ and } B) = P(A)\,P(B)$
1.5-3	Conditional probability	$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$
1.5-4	Standard deviation (sample)	$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$
1.5-5	Standard error of the mean	$\mathrm{SE} = \frac{s}{\sqrt{n}}$

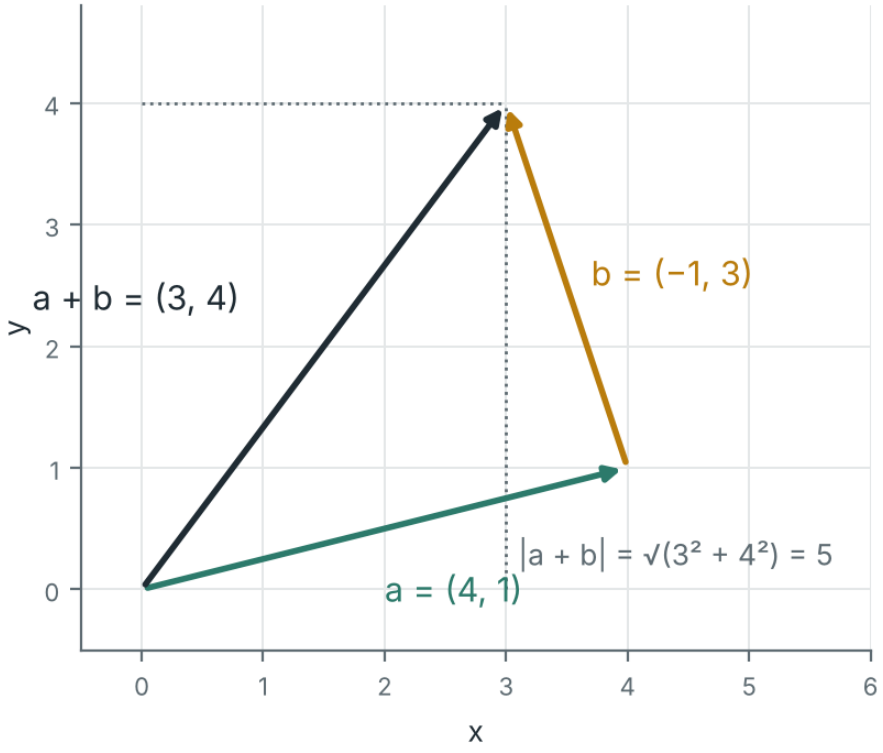


Left: the normal distribution. Right: the fraction of sixes in a growing number of simulated die rolls settles towards 1/6 (log scale on the horizontal axis).

Module 1.6 · Symbols and diagrams

Vectors

1.6-1	Magnitude	$\lvert\mathbf{a}\rvert = \sqrt{a_x^2 + a_y^2}$
1.6-2	Direction	$\theta = \tan^{-1}\left(\frac{a_y}{a_x}\right)$
1.6-3	Addition by components	$\mathbf{a} + \mathbf{b} = (a_x + b_x,\ \ a_y + b_y)$
1.6-4	Dot product, components	$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$
1.6-5	Dot product, geometric	$\mathbf{a} \cdot \mathbf{b} = \lvert\mathbf{a}\rvert\ \lvert\mathbf{b}\rvert \cos\theta$

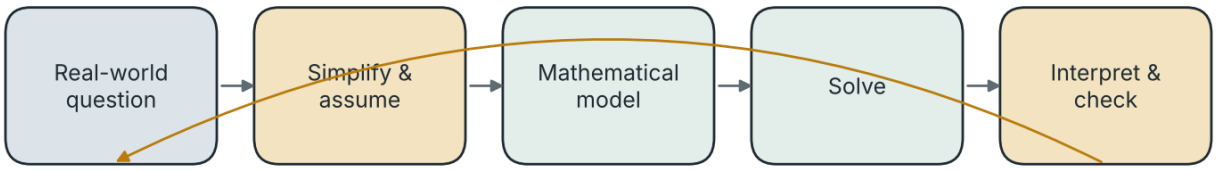


Tip-to-tail addition. Adding components gives (4, 1) + (−1, 3) = (3, 4), magnitude 5.

Module 1.7 · Symbols and diagrams

Mathematical thinking

1.7-1	Sum of two even numbers	$2m + 2n = 2(m + n)$
1.7-2	Root two, contradiction step	$\sqrt{2} = \frac{a}{b} \;\Rightarrow\; a^2 = 2b^2$
1.7-3	Sum of first n integers (proved by pairing)	$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$



The modelling cycle. The return arrow is where science happens: compare with reality and revise.